

COLUMBIA UNIVERSITY

LAMONT-DOHERTY EARTH OBSERVATORY
THE EARTH INSTITUTE AT COLUMBIA UNIVERSITY

Volcanology **I**nfrastructure for **C**omputational **T**ools and **R**esources or VICTOR, is a cloud-based cyberinfrastructure platform for volcanologists

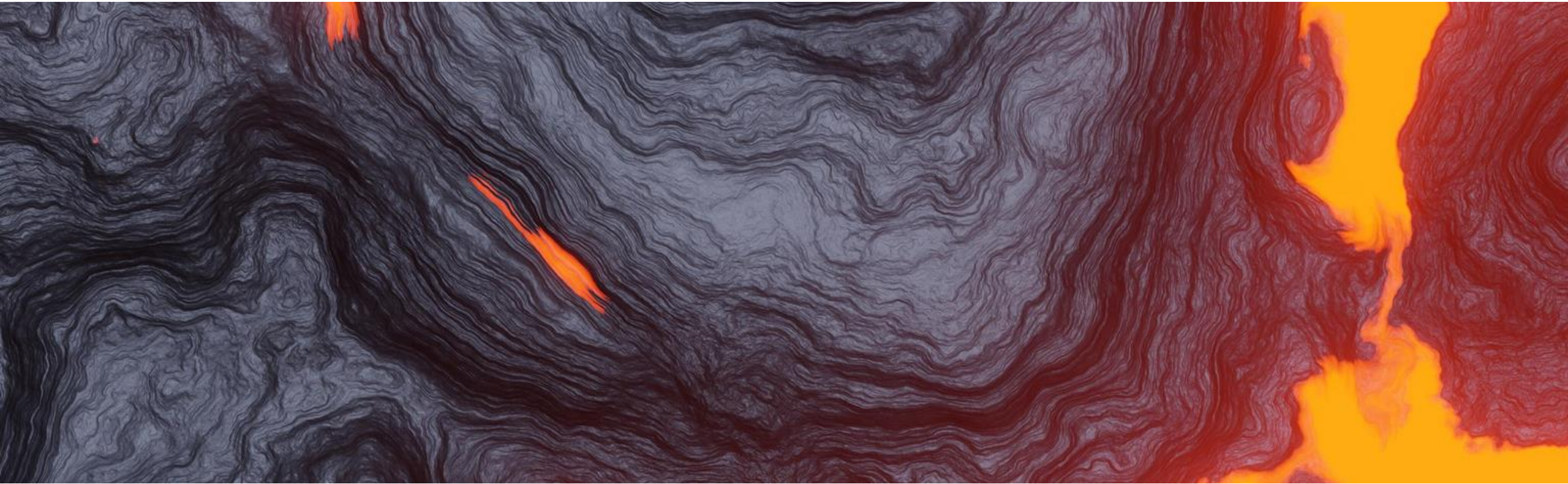


VICTOR is the successor to Vhub. However, the software and associated infrastructure are aging and inflexible, and a new medium was required

VICTOR

- VICTOR is meant to be a platform for every step of the computational process.
- Here, you can write, run, and analyze models and their associated data.
- Software libraries connect you to every corner of the cloud.

- VICTOR will help volcanologists at universities and observatories to find, use, compare, and link models of eruption processes by providing a central access portal and standardized workflows based on a JupyterHub environment.



Volcanic process/topic	Models
Tephra and plumes	Ash3d, Fall3D, Hysplit, Plumeria, Tephra2, PLUME-MoM-TSM, Puffin, Hazmap
Gas dispersal, degassing	Disgas, sulfur_X, twodee
PDC, granular flows	TITAN2D, VolcFlow, IMEX_SfloW2D
Lahars	LAHARZ, VolcFlow, IMEX_SfloW2D
Lava flows	Lava2D, MOLASSES, MrLavaLoba, Q-Lavha, Flowy, VENUSS, IMEX
Flank collapse/failure	Scoops3D
Magma thermodynamics/conduit processes	alphaMELTS, Conflow, MAGEmin, MAMMA, Perple_X, pyflowgo
Multi-hazard, Bayesian Event Tree	pyBET
Deformation tools	Magmaxisym, GNSSsphereTopoP

Launcher | molasses.ipynb

Download | Markdown | Notebook | Python 3 (ipykernel)

Run this cell and advance (Shift+Enter)

A model workflow for MOLASSES, the lava flow simulator

MOLASSES is designed to give a flow model given a set of eruption source parameter.

Three key numerical inputs must be entered

1. The lava flow residual thickness, or the thickness at rest (in meters)
2. The total volume of the effusive eruption (in m^3)
3. the lava flow of each pulse during the eruption (in m^3), which will be some fraction of the total volume.

Additionally, A DEM (Digital Elevation Model) is required for proper simulation.

Finally, an event file should be supplied.

Optionally, you can set more runs of the simulation to test/include minor variations. Keep in mind, each run will add a significant amount of runtime to the simulation.

[1]:
```
import os
import numpy as np
import rasterio as rio
from rasterio.plot import show
import subprocess
import pandas as pd
import warnings
import cartopy.crs as ccrs
import pyproj
import matplotlib.pyplot as plt
import rioxarray as rxr
import sys
warnings.filterwarnings("ignore")
%matplotlib inline
```

[2]:
```
sys.path.insert(0, "/home/jovyan/shared/Libraries/")
import victor
```

Key Parameters

Here, please input your preferred residual thickness, total volume, and pulse volume, along with the DEM filepath and events in UTM

Add

Launcher | titan2d.ipynb

Download | Code | Notebook | Python 3 (ipykernel)

A model workflow for TITAN2D

TITAN2D is a geoflow simulation tool that, given initial parameters, will simulate granular flows over digital elevation models, or DEMs.

Importing Libraries

[24]:
```
import subprocess
import pandas as pd
import numpy as np
import sys
import matplotlib.pyplot as plt
import os
import rioxarray as rxr
import rasterio as rio
import h5py
import cartopy.crs as ccrs
import glob
```

[2]:
```
sys.path.insert(0, "/home/jovyan/shared/Libraries")
import victor
```

For reference, the TITAN2D user guide is attached .

Keep in mind that for the **GIS_GRASS** type, multiple arguments to specify file mapset/map locations are neccesary, and the DEM location for the **GDAL** format. Additionally, the GDAL format requires a geotiff, rather than an ascii shapefile. We highly recommend the **GDAL** format.

There are a variety of interesting but largely unecessary files that are output at the end of the simulation. Toggle the option below to remove them.

If you would like to create a gif of all the timesteps, please enable the option below. Keep in mind, this may take many minutes, depending on the number of iterations.

[3]:
```
cleanup = False

create_gif = True
```

More resources about VICTOR

- https://victor.ldeo.columbia.edu/
- https://docs.victorproject.org/
- https://www.youtube.com/watch?v=3wgfqFVSYwk

VICTOR set-up and course files

- https://hub.victorproject.org/



Operated by: **2i2c** | Funded by: **National Science Foundation** | Designed by: **2i2c**

Log in to continue

Selected Identity Provider

GitHub

Remember this selection

Log On

By selecting "Log On", you agree to the privacy policy.

Server Options

CPU Only

Image: Victor Notebook

Resource Allocation: 3.7 GB RAM, upto 3.7 CPUs

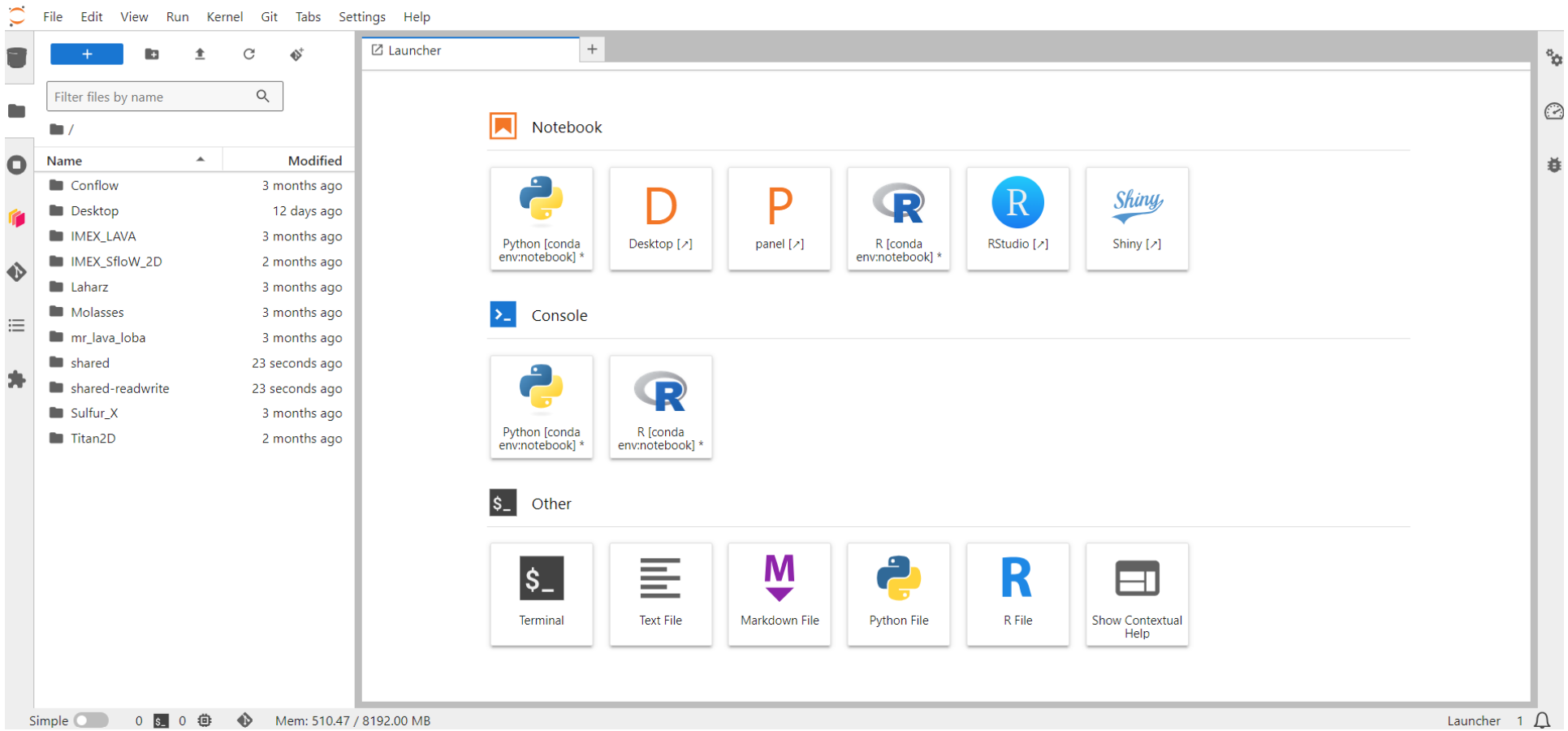
- 3.7 GB RAM, upto 3.7 CPUs
- 7.4 GB RAM, upto 3.7 CPUs
- 14.8 GB RAM, upto 3.7 CPUs
- 29.7 GB RAM, upto 3.7 CPUs
- 32.5 GB RAM, upto 7.8 CPUs
- 60.6 GB RAM, upto 15.6 CPUs

NVIDIA Tesl

Start a container on a dec

Image

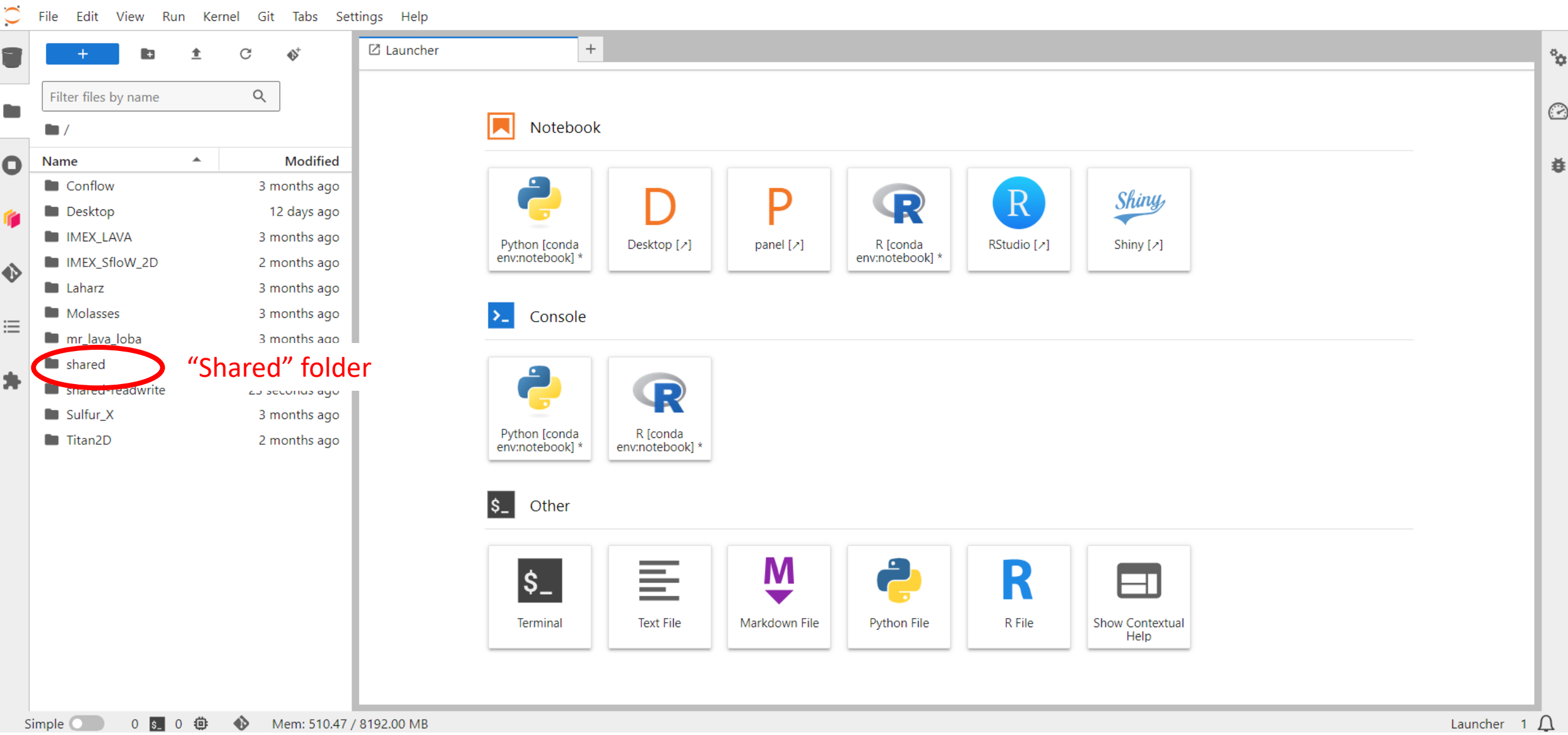
Start

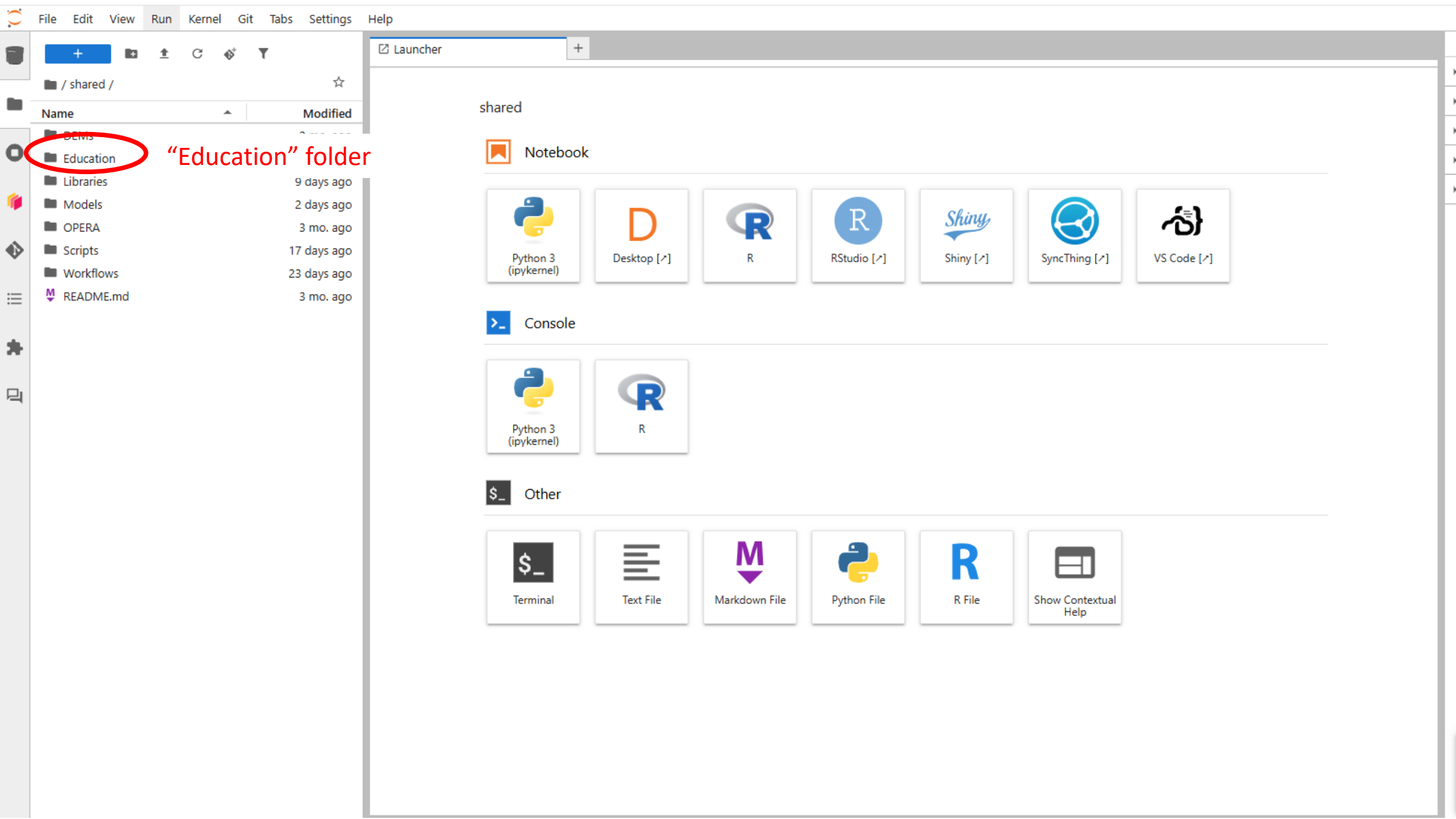


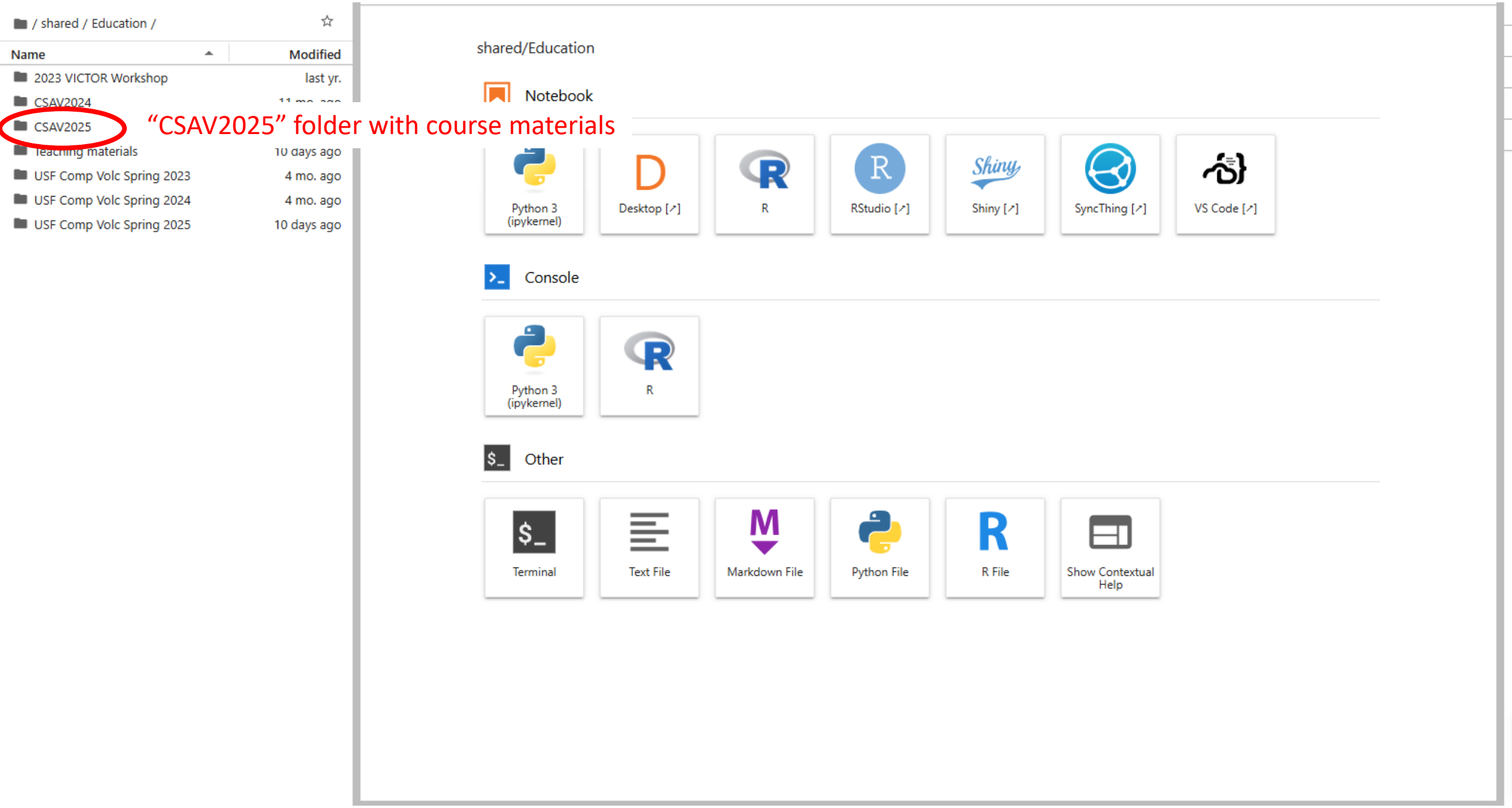
- When you login to VICTOR, your screen will look something like this
- Your files are on the left panel
- Your “Launcher” has useful icons

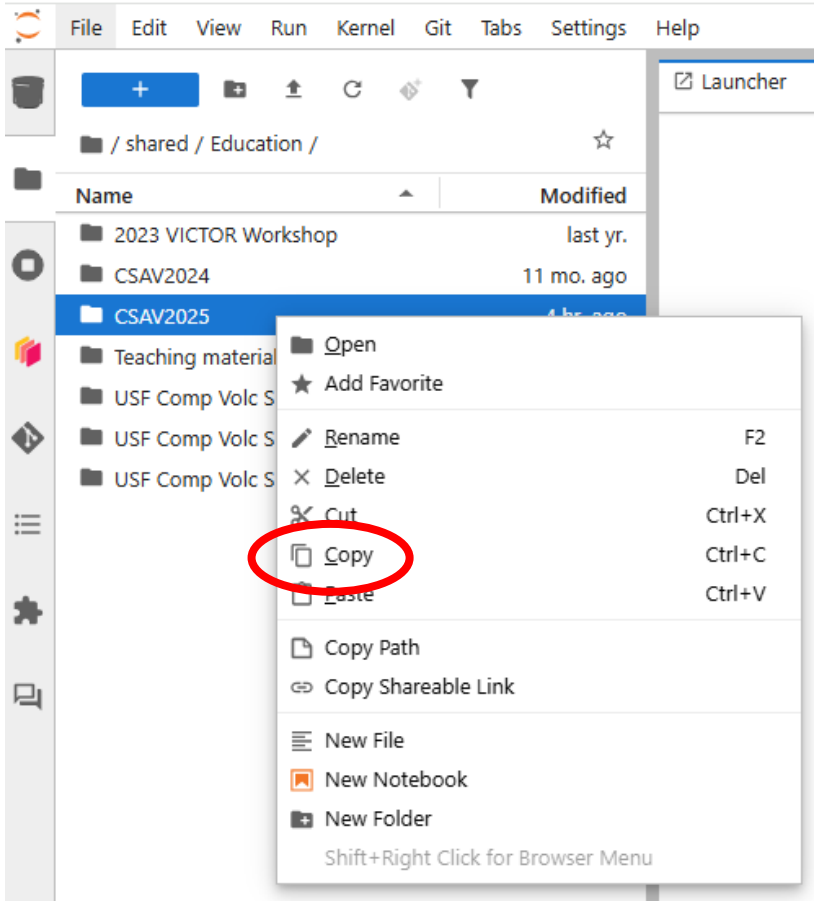
Tip: If you lose your “Launcher” screen, click the “plus” icon at the top left for “New Launcher”

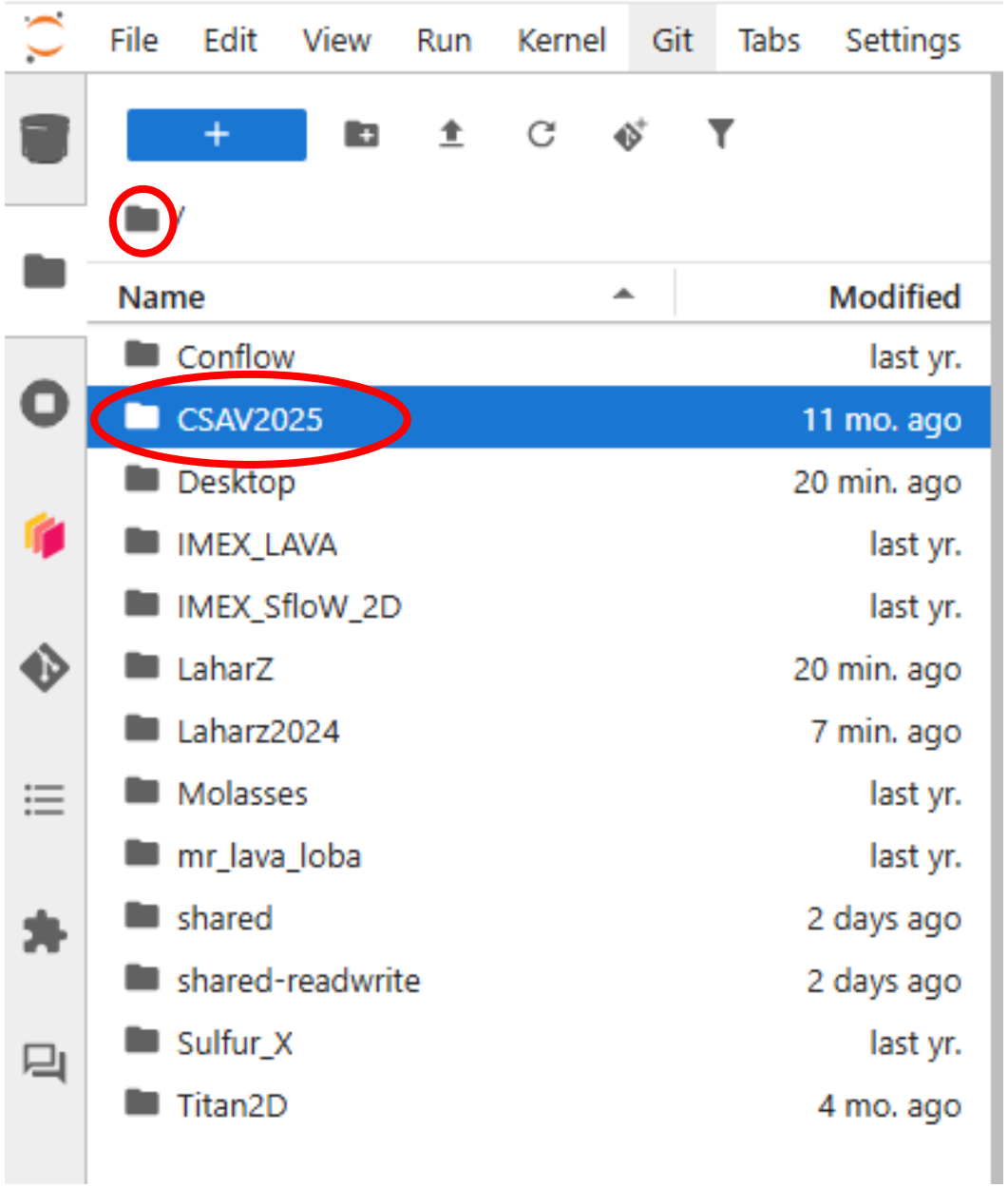
Our course files are in /shared/Education/CSAV2025



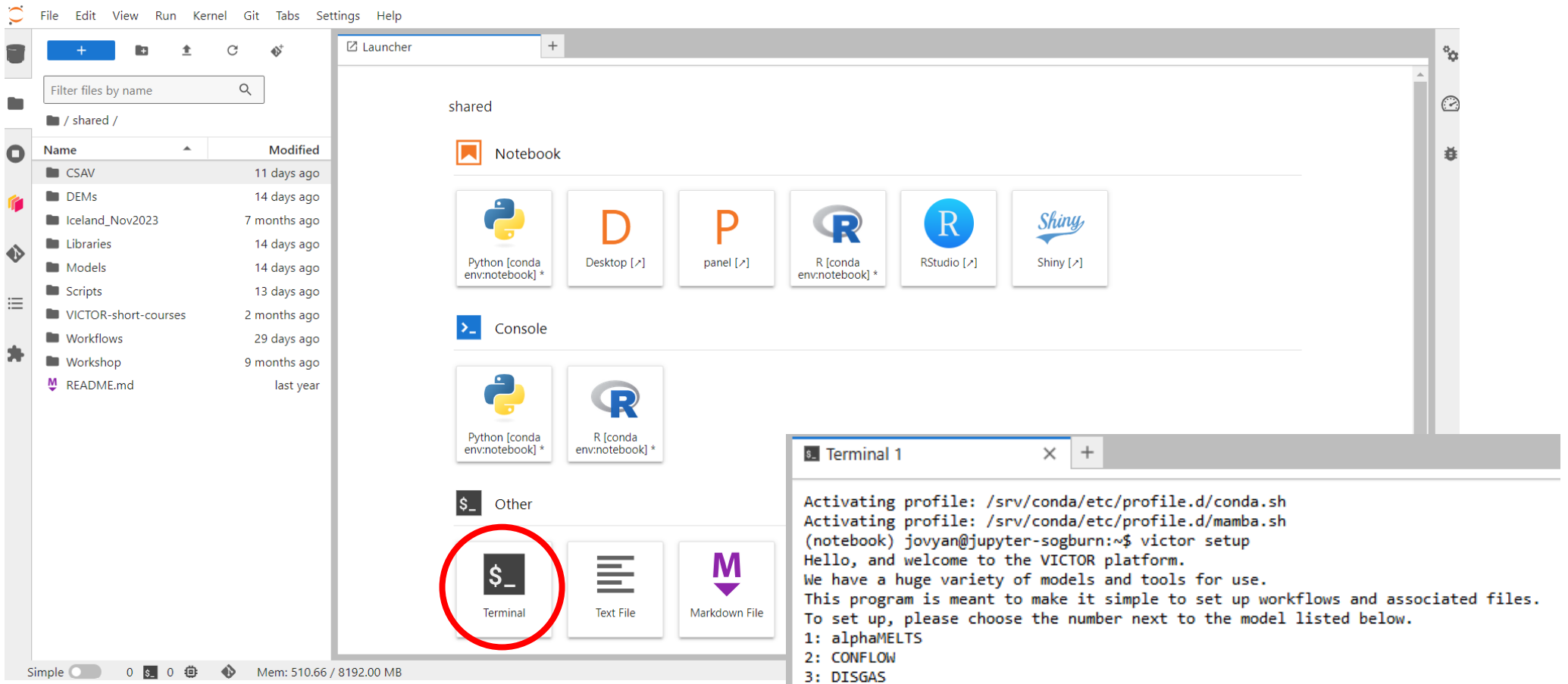

File
Edit
View
Run
Kernel
Git
Tabs
Settings
Help
/ shared /
Name
Modified
Education
Libraries
9 days ago
Models
2 days ago
OPERA
3 mo. ago
Scripts
17 days ago
Workflows
23 days ago
README.md
3 mo. ago
“Education” folder
Launcher
shared
Notebook
Python 3 (ipykernel)
Desktop [↗]
R
RStudio [↗]
Shiny [↗]
SyncThing [↗]
VS Code [↗]
Console
Python 3 (ipykernel)
R
Other
Terminal
Text File
Markdown File
Python File
R File
Show Contextual Help

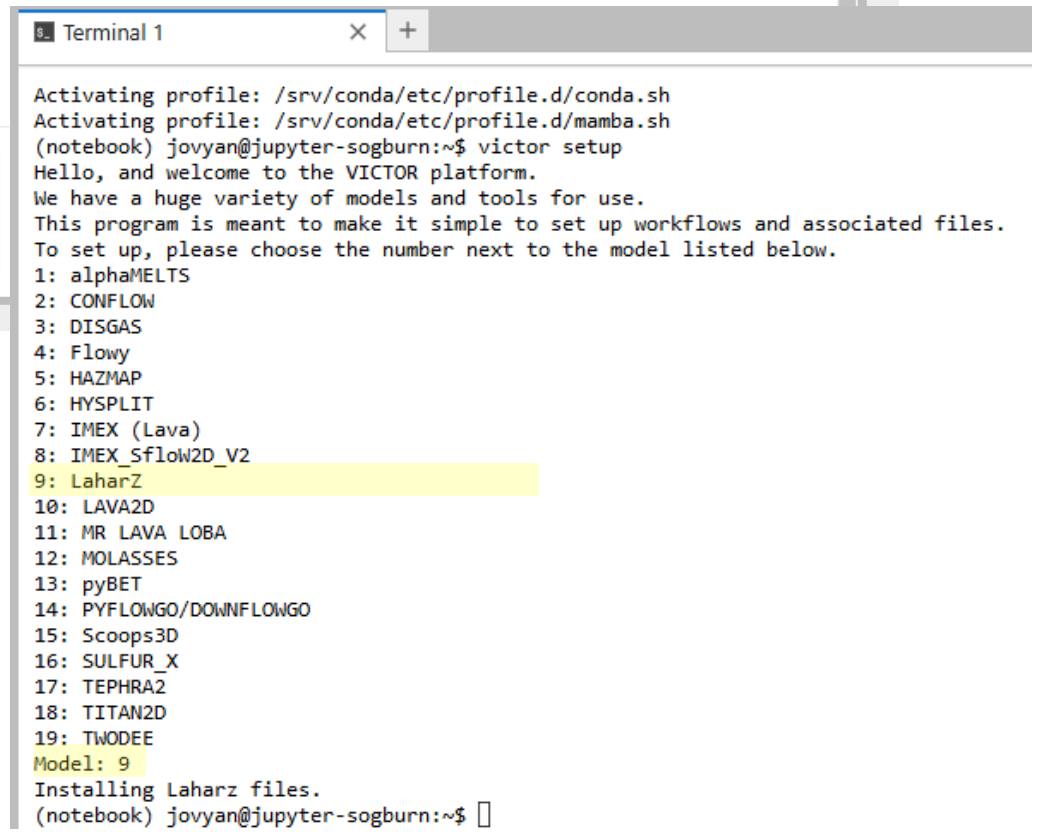

/ shared / Education /
Name
Modified
2023 VICTOR Workshop
last yr.
CSAV2024
CSAV2025
Teaching materials
10 days ago
USF Comp Volc Spring 2023
4 mo. ago
USF Comp Volc Spring 2024
4 mo. ago
USF Comp Volc Spring 2025
10 days ago
"CSAV2025" folder with course materials
shared/Education
Notebook
Python 3 (ipykernel)
Desktop [↗]
R
RStudio [↗]
Shiny [↗]
SyncThing [↗]
VS Code [↗]
Console
Python 3 (ipykernel)
R
Other
Terminal
Text File
Markdown File
Python File
R File
Show Contextual Help





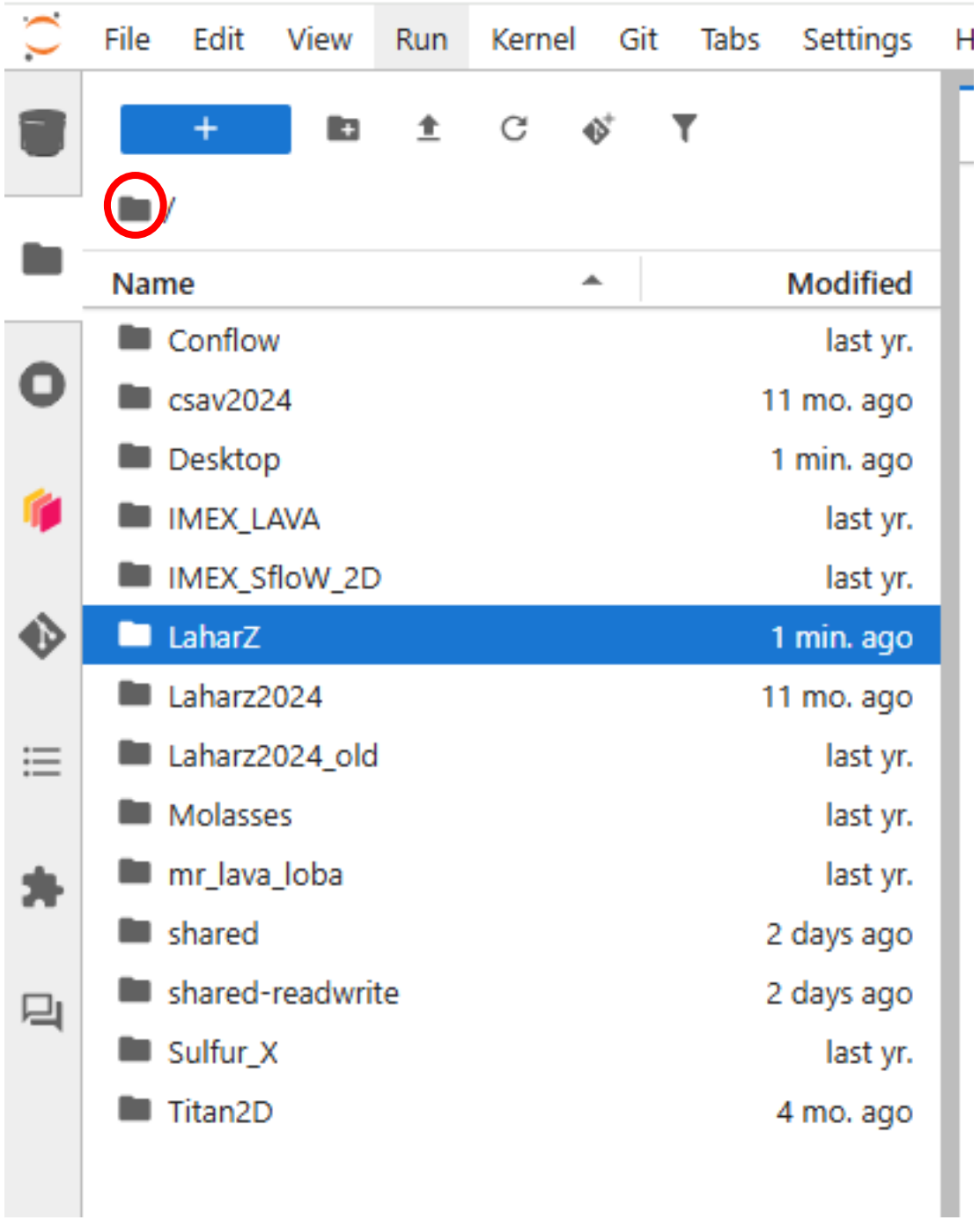
- Right click on the “CSAV2025” folder
- Click “Copy”
- Navigate to your home folder (click the folder icon)
- Paste the “CSAV2025” folder there

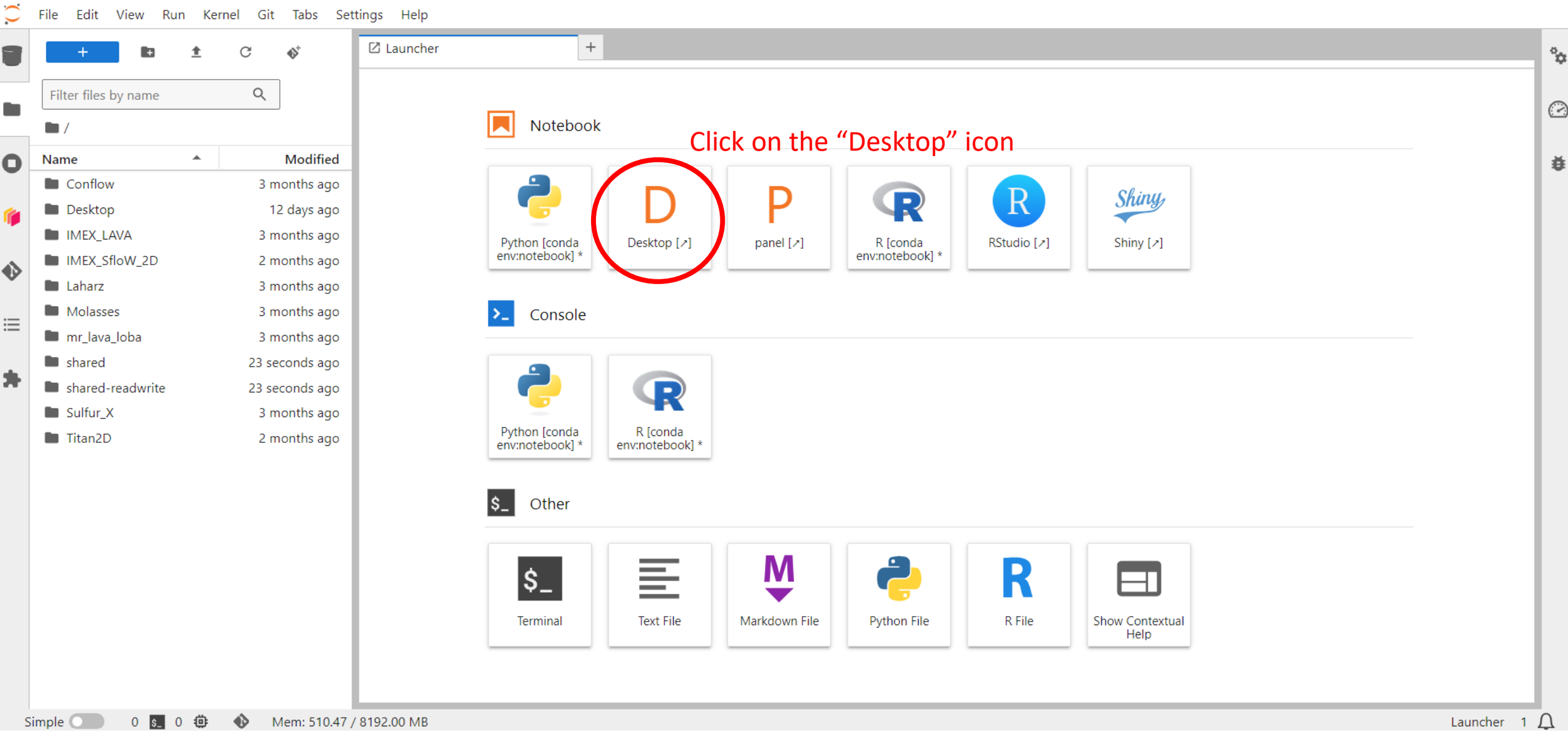


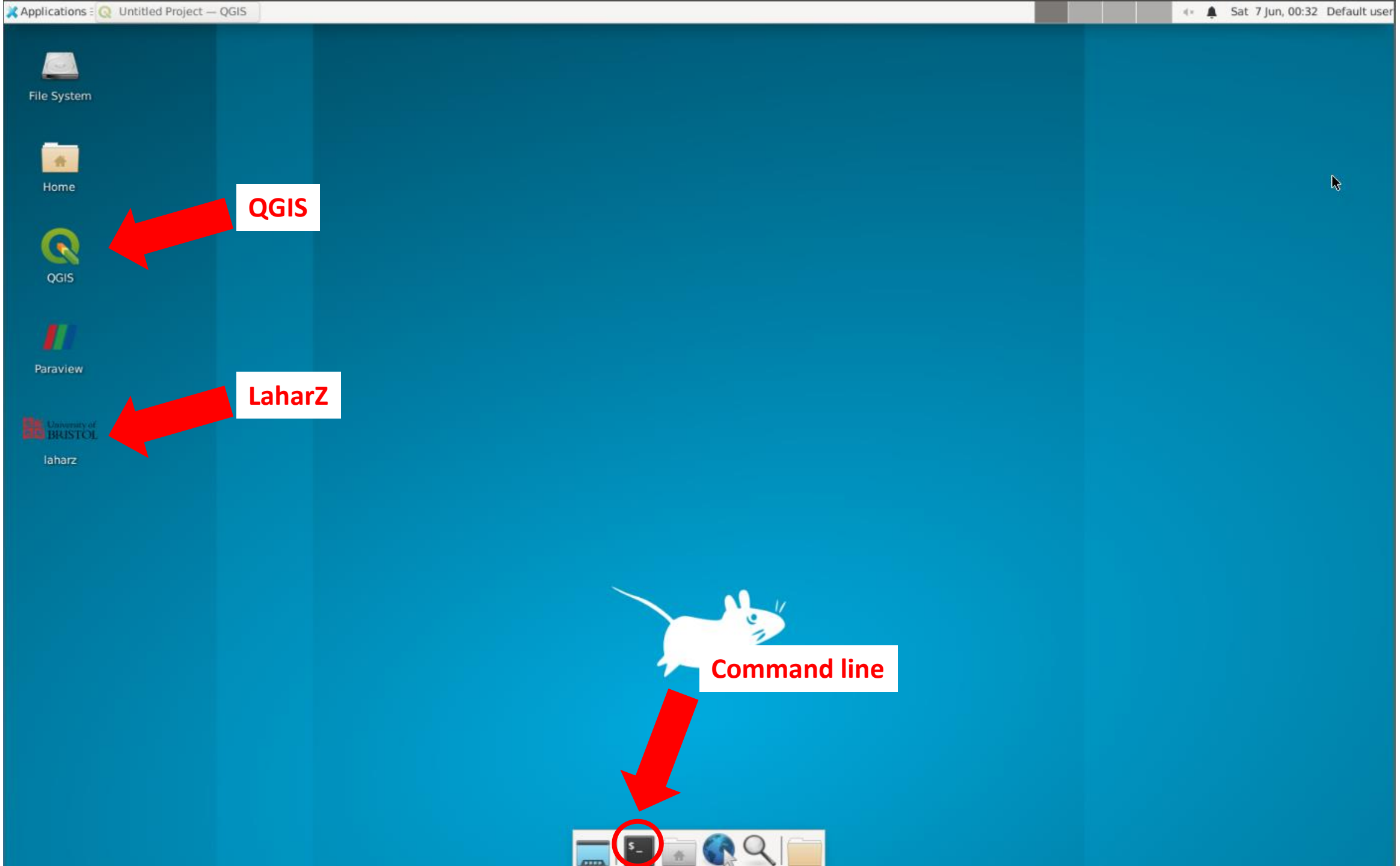


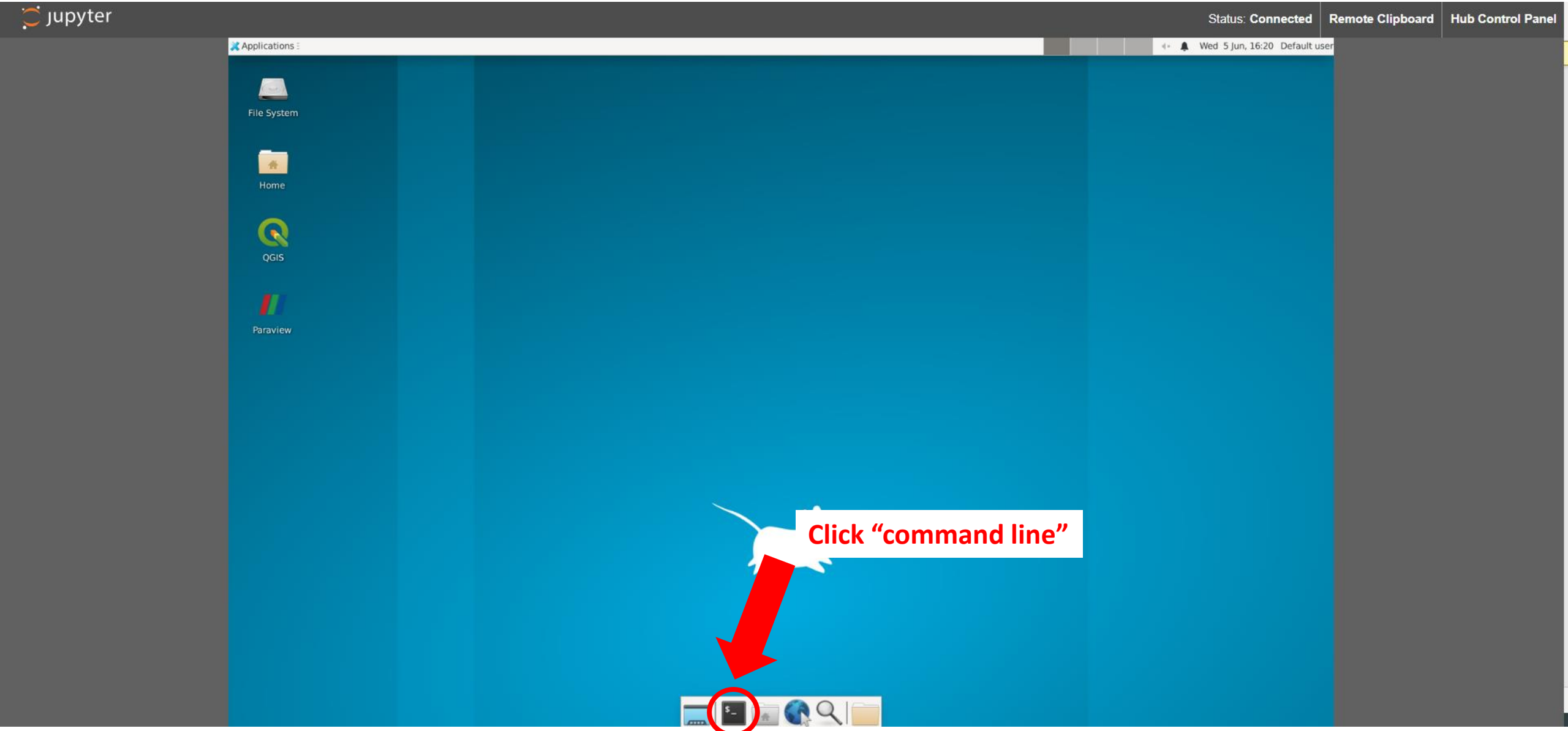
- We want to add the LaharZ model to your home workspace
- Click the terminal
- Type: victor setup
- Hit enter on your keyboard
- Type: 9
- Hit enter
- Close the terminal

- Check to see if the LaharZ folder was created in your space
- Click the home folder and look for the LaharZ folder (You will have different files than me, don’t worry!)



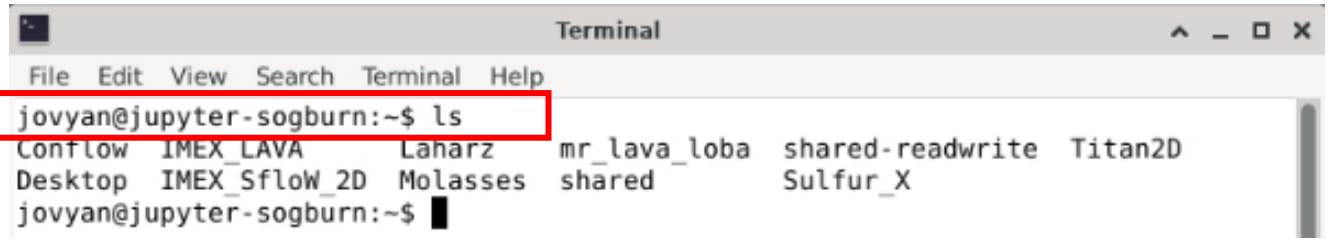

File Edit View Run Kernel Git Tabs Settings Help
Launcher
Filter files by name
/
Name
Modified
Conflow 3 months ago
Desktop 12 days ago
IMEX_LAVA 3 months ago
IMEX_SfloW_2D 2 months ago
Laharz 3 months ago
Molasses 3 months ago
mr_lava_loba 3 months ago
shared 23 seconds ago
shared-readwrite 23 seconds ago
Sulfur_X 3 months ago
Titan2D 2 months ago
Notebook
Click on the “Desktop” icon
Python [conda env:notebook] *
Desktop [↗]
panel [↗]
R [conda env:notebook] *
RStudio [↗]
Shiny [↗]
Console
Python [conda env:notebook] *
R [conda env:notebook] *
Other
Terminal
Text File
Markdown File
Python File
R File
Show Contextual Help
Simple
0
0
Mem: 510.47 / 8192.00 MB
Launcher
1

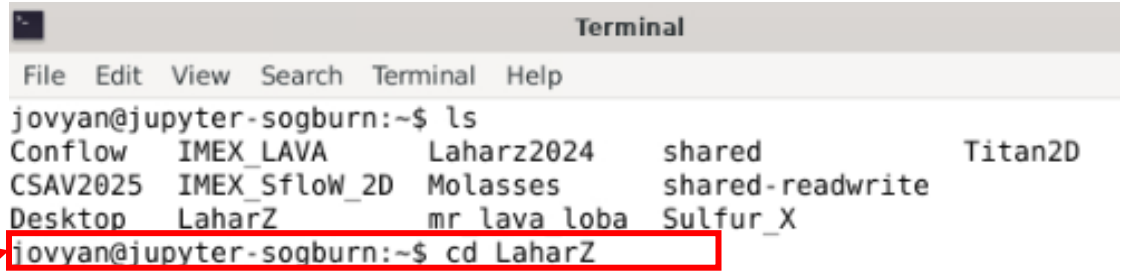

Applications
Untitled Project — QGIS
Sat 7 Jun, 00:32
Default user
File System
Home
QGIS
QGIS
Paraview
LaharZ
laharz
Command line

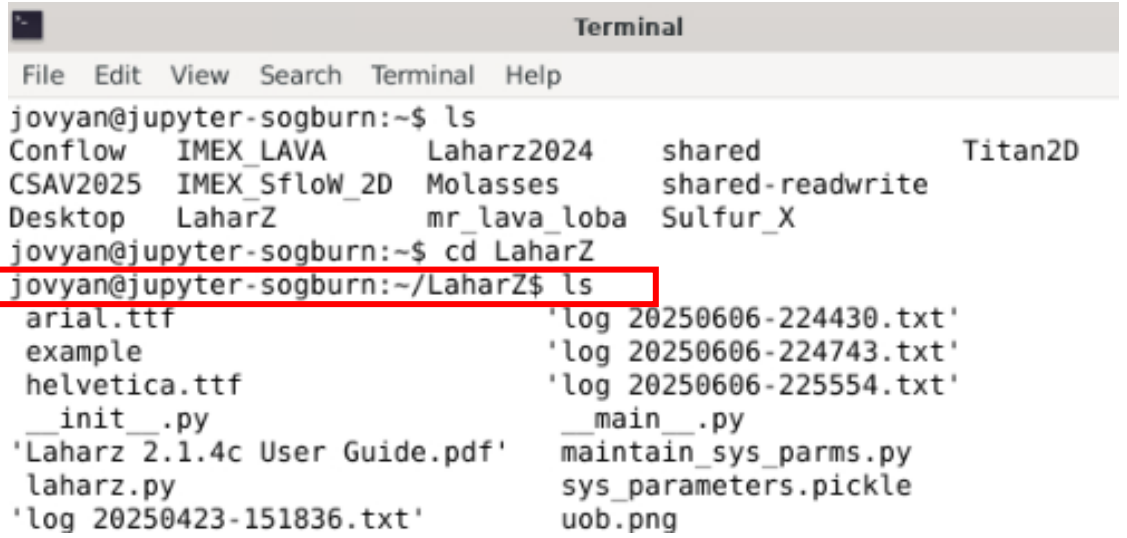

jupyter
Status: Connected
Remote Clipboard
Hub Control Panel
Applications
Wed 5 Jun, 16:20 Default user
File System
Home
QGIS
Paraview
Click “command line”

Using the command line

- Type a command and hit “enter”
- To list the files in the folder you’re in: **ls**
- To move to a folder: cd name_of_folder
  - If you type part of a folder name and then click the “tab” key, it will autocomplete
  - Move to LaharZ folder: **cd LaharZ/**
- List what’s in the “LaharZ” folder: **ls**
  - You’ll see a user manual and the lahars.py code along with some other files

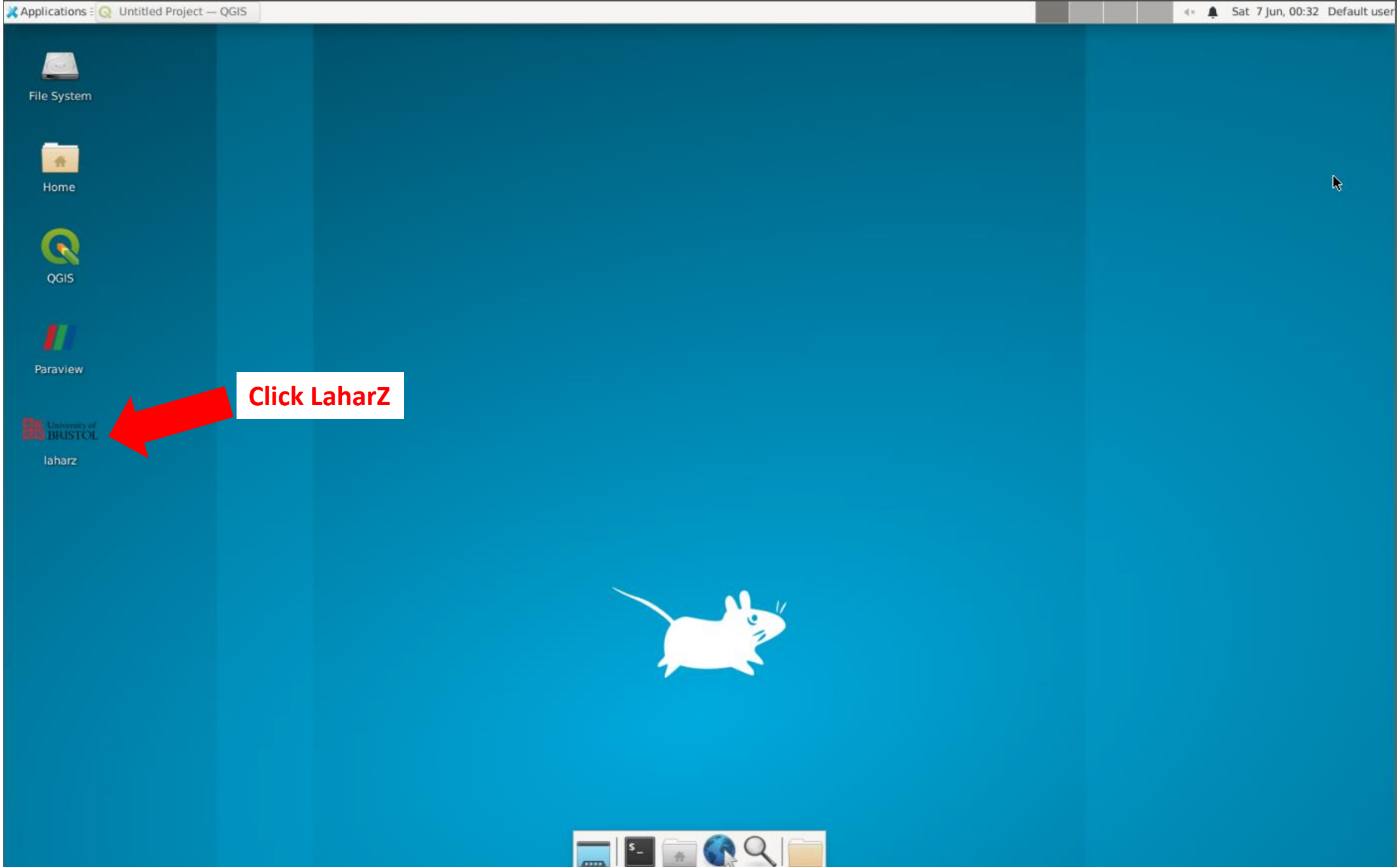






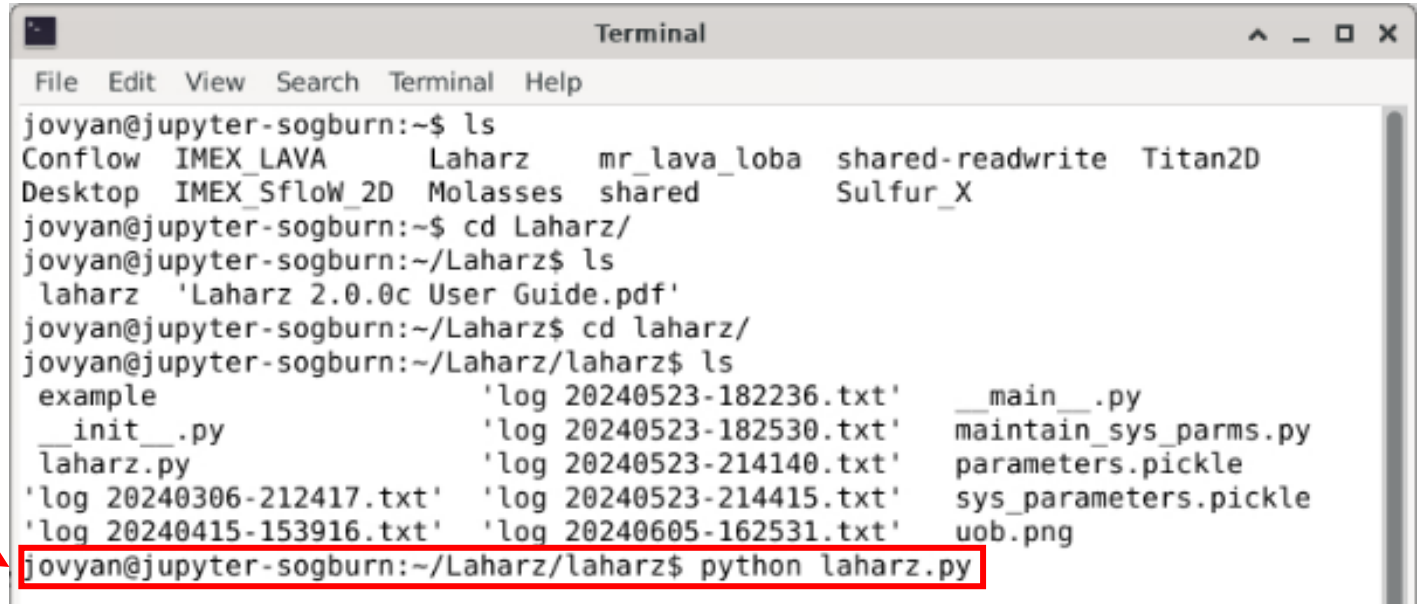
Some command line tips

- If you type part of a folder name and then click the “tab” key, it will autocomplete
- Clicking the up arrow on the keyboard will cycle through commands you’ve typed before
- File/Hub Control Panel/Stop Server if needed

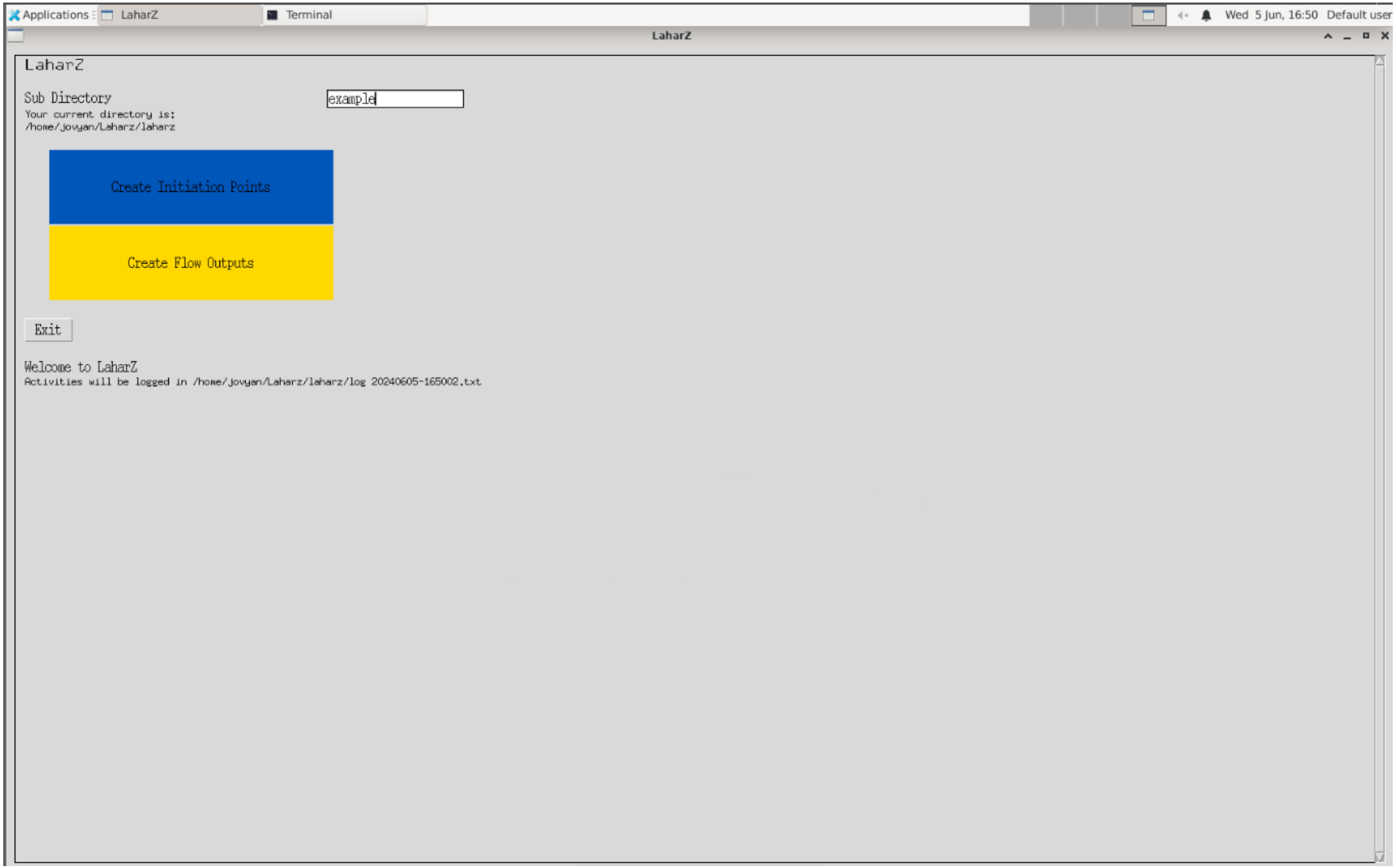

Applications
Untitled Project — QGIS
Sat 7 Jun, 00:32 Default user
File System
Home
QGIS
Paraview
Click LaharZ
laharz

Launch LAHARZ

- Launch LAHARZ: **python laharz.py**




Applications
LaharZ
Terminal
Wed 5 Jun, 16:50 Default user
LaharZ
LaharZ
University of
BRISTOL
Version: 2.0.0 stand alone


Applications
LaharZ
Terminal
Wed 5 Jun, 16:50 Default user
LaharZ
LaharZ
Sub Directory
example
Your current directory is:
/home/jovyan/Laharz/laharz
Create Initiation Points
Create Flow Outputs
Exit
Welcome to LaharZ
Activities will be logged in /home/jovyan/Laharz/laharz/log 20240605-165002.txt